THIAGARAJAR COLLEGE OF ENGINEERING, MADURAI 625 015 | Department of EEE

AGROVOLTAIC CANOPY FOR PROTECTION AND SUSTAINABLE FARMING

Six sensors in, best panel angle out: the crop and the solar panels both win

Thanwin S S | Chandru P M | Kuberaeshwaran S | Mukilan A Guide: Dr. M. Meenakshi Devi

REVIEW 0
Project III | 2026-27

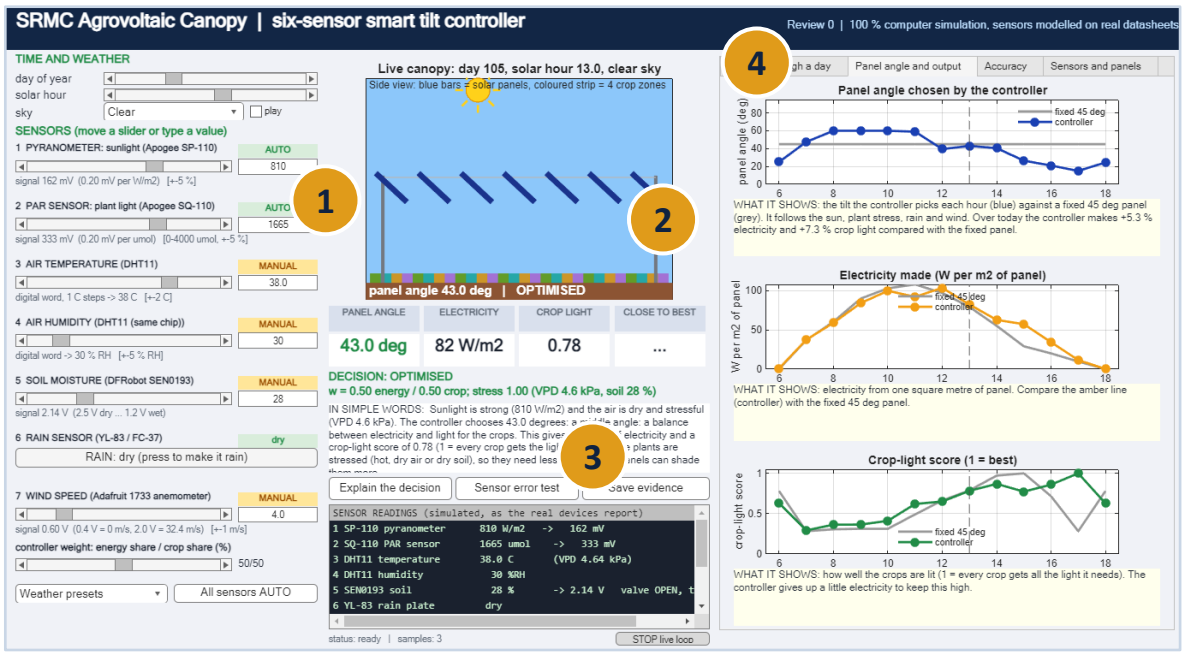
99.6 %
accuracy of the co-optimisation against the best possible angle (fixed 45 deg: 91.2 %)

6
sensors, each simulated from a real datasheet and adjustable by hand

4.4 kWp
8 x JinkoSolar Tiger Pro JKM550M-72HL4-V panels, 550 Wp each

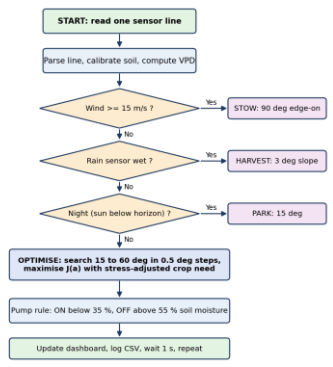
29 / 29
software tests pass; sunlight checked against NASA POWER and PVGIS

1 LIVE MATLAB DASHBOARD: change any sensor, the panels react



1 Seven readings from six real devices: slider or typed number, tagged MANUAL or AUTO. **2** Live canopy: the blue panels tilt.
3 The decision, in plain words. **4** Day graphs: sensors, angle, electricity, crop light, accuracy. Each graph has a one-line explanation.

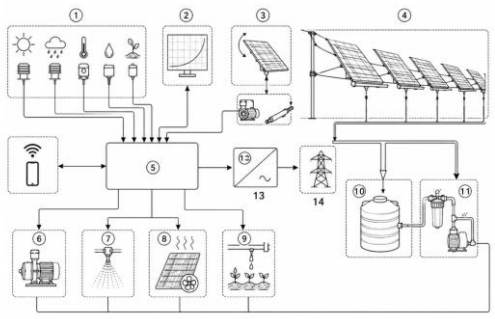
2 HOW IT DECIDES



Safety first
Wind 15 m/s or more: stow at 90 deg
Rain: 3 deg, fill the tank
Night: park at 15 deg

Otherwise
Best of 15-60 deg for 1/2 electricity + 1/2 crop light; more shade if hot and dry

3 THE SYSTEM



4 SIX SENSORS = SIX REAL DEVICES

Apogee SP-110
pyranometer: sunlight
0-2000 W/m2, +-5 %

Apogee SQ-110
PAR: light plants use
0-4000 umol/m2/s, +-5 %

DHT11
air temperature and humidity
0-50 C +-2 C | 20-90 % +-5 %

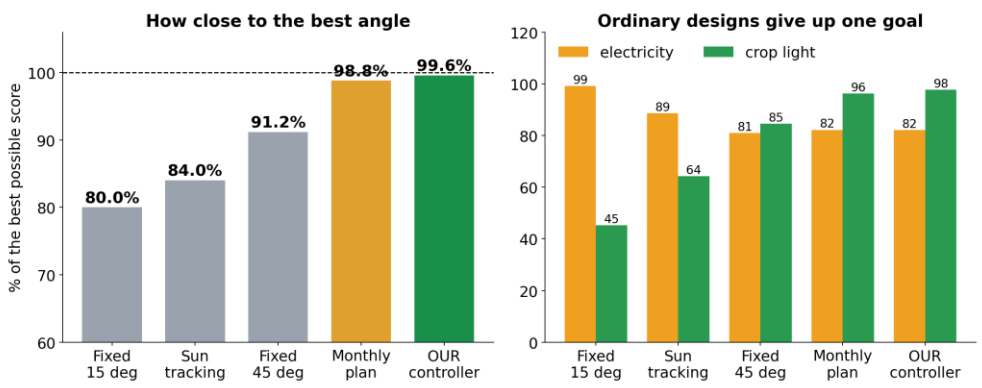
DFRobot SEN0193
soil moisture (capacitive)
1.2-2.5 V analog

YL-83 / FC-37
rain plate + LM393
wet / dry, digital

Adafruit 1733
cup anemometer: wind
0-32.4 m/s, +-1 m/s

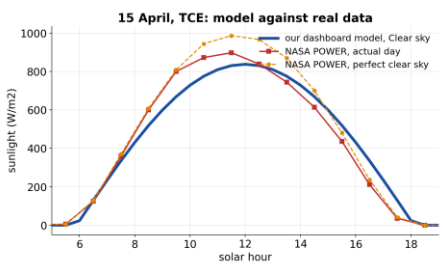
Panels: 8 x JinkoSolar Tiger Pro JKM550M-72HL4-V, 550 Wp, 40.90 V, 13.45 A, 21.33 %, 2274 x 1134 mm (datasheet on file)

5 ACCURACY: WHAT WE OBTAINED



Our six-sensor controller: 99.6 % of the best possible score, 4.8 deg from the best angle. Fixed 45 deg: 91.2 % (16.3 deg). Sun tracking: 84.0 %. Fixed 15 deg: 80.0 %. *864 noisy cases, simulation.*

6 CAN YOU TRUST THE SUNLIGHT?



Dashboard Clear day 6502 Wh/m2 vs NASA POWER 6539 and PVGIS 6337. Plant-light share 0.444 vs 0.45. A check against satellite data, not a roof sensor.

7 PROOF YOU CAN OPEN

- ✓ 7 sensor and panel datasheets, downloaded
- ✓ NASA POWER + PVGIS sunlight data
- ✓ Model vs data figure and numbers
- ✓ Accuracy test file (864 cases)
- ✓ SHA-256 checksum of every proof file
- ✓ Evidence Book (Word) for the judges
- ✓ Code and results on GitHub

8 IF IT WERE REAL: COST (INR)

Research-grade sensing kit	**67,600**
Low-cost sensing kit	**13,100**
8 panels, 4.4 kWp (assumed)	**70,000**
This simulation	**0**

Estimates, not quotes. Sensors: 2 Apogee ~INR 56,000, anemometer 6,500, DHT11 + soil + rain 700, ESP32, ADC, wiring.

NEXT Real sensors and one month of logged data | wind-tunnel or CFD run to settle the storm stow | crop-light and yield measurement | paper submission

HONEST LIMITS Simulation only. Sensor readings are generated or typed, not measured. Accuracy is against the controller's own goal, not crop yield. Prices are estimates.



Scan for code and proof